

Post-quantum cryptography readiness assessments and migration roadmaps for EU financial institutions to meet DORA and NIS2 third-party cyber risk management requirements.

Finance, Banking and Insurance × Regulatory compliance and reporting × Post-quantum cryptography · OS138 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
72 is the world moving	48 can we play, can we win	MEDIUM 1 named in the evidence	€189m modelled evidence · per year	NEXT regulation adopted or propose...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 4.0 published reference(s) in Finance, Banking and Insurance, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is a post-quantum cryptography (PQC) readiness assessment and migration roadmap service for EU financial institutions, addressing the cryptographic readiness requirements embedded in DORA and NIS2 third-party cyber risk management. The service helps banks and insurers identify where quantum-vulnerable cryptography exists across their estate, prioritize remediation, and plan a compliant migration. It exists now because NIST has finalized PQC standards, regulators are enforcing third-party risk obligations, and enterprise adoption remains critically low.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€231m €41.6m – €1.1bn	€189m €33.9m – €907m	€2.3m €407k – €10.9m	modelled
Observed: tendered contract value, annualised	€100m	€100m	€1.2m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Finance, Banking and Insurance (EU27_2020)	11,116 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Post-quantum cryptography	39.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cisce_ran2 · E_SECMDEC	2024
Annual value of one engagement	€240k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	2,426 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. ICT security measures by NACE Rev. 2 activity series E_SECMDEC is used as a proxy for Post-quantum cryptography: Encryption users are the migration population. Part of this vertical is not covered separately by the enterprise ICT survey, so the all-activities rate was used for those slices. Contract value is a configured assumption, not observed. Only 2 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. Sector adoption is read from Eurostat's all-activities aggregate for part of this vertical, which the enterprise ICT survey does not cover separately: PROXY — ICT survey excludes the financial sector; PROXY — auxiliary financial activities; PROXY — insurance and pension funding

Observed: tendered contract value, annualised

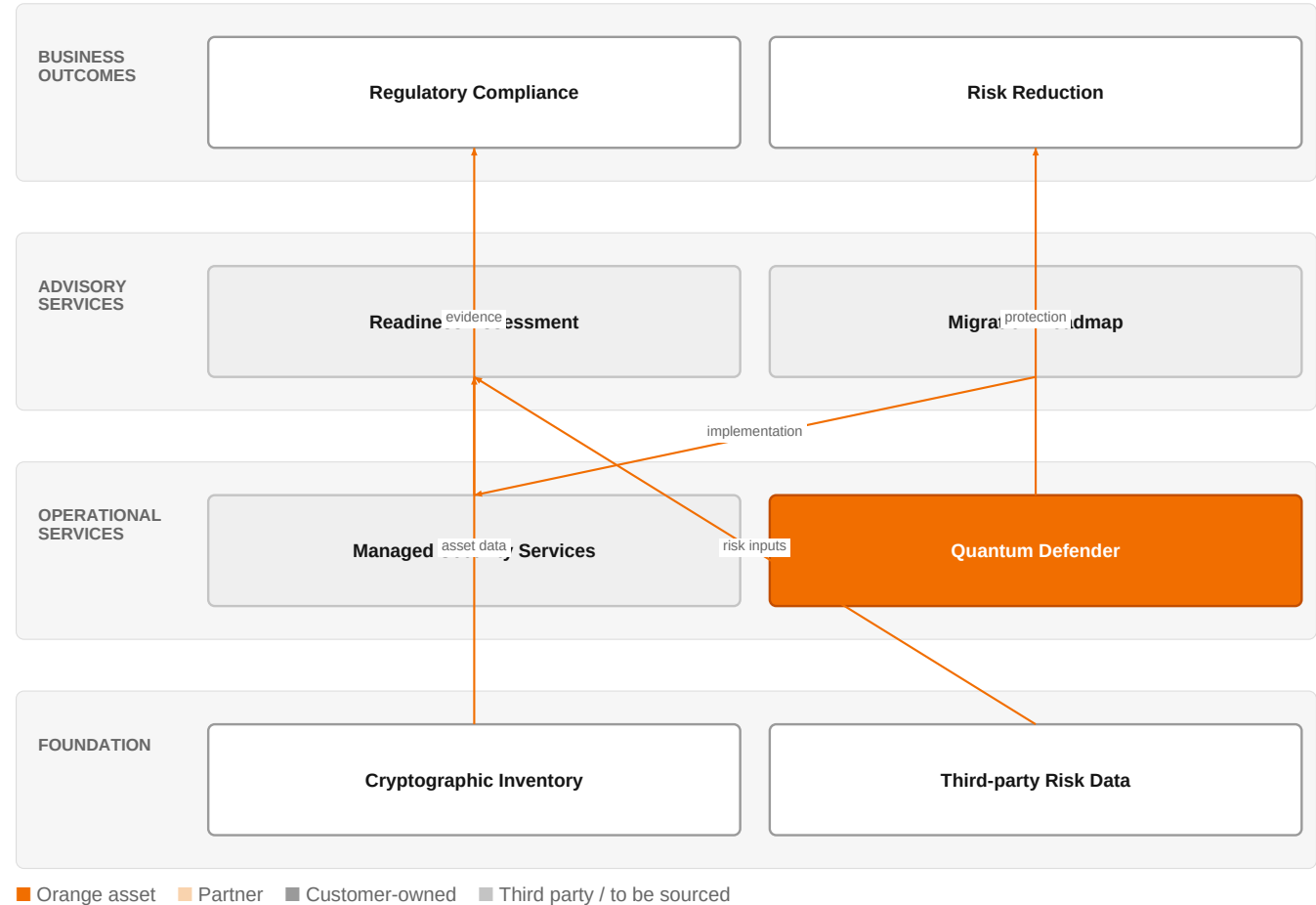
Factor	Value	Basis	Source	As of
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Tendered contract value in matching CPV categories	€100m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

PQC Readiness and Migration Roadmap



This diagram shows how Orange's advisory and managed services turn a customer's existing cryptographic inventory and third-party risk data into a compliant, risk-reduced PQC migration.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

NIST has finalized post-quantum cryptography standards (FIPS 203, 204, 205), triggering an unprecedented cryptographic migration, yet enterprise adoption remains critically low. DORA and NIS2 impose requirements on third-party cyber risk management, which can include cryptographic readiness. Migration is a multi-year, multi-domain transformation across cloud, enterprise, embedded, and OT systems, requiring ecosystem coordination and larger parameter sizes. Regulators and standardization bodies have launched initiatives, but a critical gap remains: operational models to quantify, compare, and prioritize asset exposure.

Sources: SIG-B5F1E5C72CAB Springer Science and B 2026-07-30; SIG-9C40ACE674BC Wolters Kluwer 2026-08-10; SIG-57F4F0DAE044 Cryptography 2026-06-18; SIG-8BD0B85D3181 openalex.org 2025-12-24; SIG-D45704585F54 Elsevier BV 2026-01-01

Who buys, and why now

The Chief Information Security Officer (CISO) or Head of Cyber Risk signs the engagement, driven by the second line of defense (risk, compliance, audit) which must ensure the migration is auditable and aligned with regulator expectations. The pain is operational: they cannot inventory cryptographic assets across complex estates, especially embedded systems and third-party dependencies, and they lack a risk-based prioritization framework. The trigger is an upcoming

DORA or NIS2 compliance audit, or a regulator inquiry about quantum readiness.

Sources: SIG-5C7568EBD975 Elsevier BV 2026-01-01; SIG-13AAE2CD702E MDPI AG 2026-01-19; SIG-D45704585F54 Elsevier BV 2026-01-01

Why it is hot now — every claim bound to a dated source

- DORA and NIS2 impose requirements on third-party cyber risk management, which can include cryptographic readiness. [SIG-9C40ACE674BC]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a PQC readiness assessment and migration roadmap using its Cybersecurity experts and Defense and security experts, supported by Orange Group researchers. The engagement starts with a cryptographic asset discovery and inventory across the customer's estate, leveraging Orange Cyberdefense managed services for continuous monitoring. Next, a risk-based prioritization framework (quantum risk index) is applied to identify critical exposures. Finally, a phased migration roadmap is delivered, aligned with DORA/NIS2 third-party risk requirements, and optionally supported by Quantum Defender / quantum-safe offers for implementation.

Why Orange can win

Orange has a credible combination of cybersecurity expertise, defense-grade security knowledge, and research capability, all backed by ISO 27001, SOC 2 Type II, and CISPE certifications. The ability to leverage Orange Cyberdefense managed services provides a continuous, operational angle beyond a one-off assessment. However, the assets are thin on specific PQC migration tooling and quantum-safe product depth, so Orange must lead with its governance and risk framework rather than point tools.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Quantum Defender / quantum-safe offers	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SOC 2 Type II	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
CISPE	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Banqsoft	reference	SUP	published reference in this vertical, different use case	unconfirmed
Stø	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.4; 1 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include global systems integrators like Atos/Eviden, IBM, and Accenture, who sell PQC migration services and have strong finance sector presence. Security specialists such as Airbus Protect, Entrust, ID Quantique, and Thales offer PQC-focused products and services. Microsoft, as a hyperscaler and Orange partner, is also active in this space. The customer will compare Orange's ability to integrate with their existing infrastructure and provide a sovereign, certified delivery against these alternatives.

Sources: SIG-5C7568EBD975 Elsevier BV 2026-01-01

Competitor	Category	Position here	Basis	Evidence
Microsoft	Hyperscaler	named in this topic's evidence — also an Orange partner	evidenced	SIG-5C7568EBD975
Atos / Eviden	Global systems integrator	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
IBM	Global systems integrator	sells Post-quantum cryptography; active in Finance, Banking and Insurance; competes in Cybersecurity	structural	—
Airbus Protect	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Entrust	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
ID Quantique	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
Thales	Security specialist	sells Post-quantum cryptography; active in Finance, Banking and Insurance; competes in Cybersecurity	structural	—
Accenture	Global systems integrator	active in Finance, Banking and Insurance	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Have you completed a cryptographic asset inventory across your entire estate, including third-party and embedded systems?
- 2 What is your current timeline for DORA or NIS2 compliance audits, and has quantum readiness been raised by your second line of defense?
- 3 Do you have a risk-based framework to prioritize which cryptographic assets to migrate first?
- 4 Have you identified which of your third-party providers use quantum-vulnerable cryptography, and do you have contractual clauses to address this?
- 5 What is your current level of awareness of NIST FIPS 203/204/205 and how they apply to your infrastructure?

Objections you will meet

They will say	You can answer
We already have a cybersecurity team and don't need external help.	That's fair, but PQC migration is a specialized, multi-year effort that requires dedicated expertise. Our assessment can complement your team by providing an independent, risk-based view and a roadmap that aligns with DORA/NIS2, which your internal team may not have the bandwidth to develop.
Quantum computers are not a near-term threat, so we can wait.	You're right that the threat is not imminent, but migration timelines are long and the effort is substantial. Starting now allows you to plan and budget, and regulators are increasingly expecting proactive readiness. Our assessment helps you understand your exposure and create a realistic roadmap without disrupting current operations.
We already have a preferred vendor for cybersecurity services.	That's understandable. However, our approach is complementary: we can work alongside your existing vendors, focusing specifically on the PQC readiness and regulatory compliance aspects. Our certifications and defense-grade expertise add a layer of assurance that may not be covered by your current provider.

Risks and unknowns

The main risk is that PQC migration is a multi-year effort and customers may defer action until regulators enforce deadlines. There is also uncertainty about the actual quantum threat timeline and the interoperability of hybrid schemes. Orange's specific PQC tooling and methodology are not fully proven in the market, and the competitive landscape is crowded with established players. The lack of a named reference in this exact domain is a gap.

Next action, by role (FR-17)

Role	Do this next
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Presales / Proposal	Prepare a bid brief for a European financial institution, leveraging the Quantum Defender offer and Orange Cyberdefense managed services, and referencing our ISO 27001 and SOC 2 Type II certifications to address DORA and NIS2 third-party risk requirements.
Sales	In a meeting with a European bank's CISO, say: 'Given the upcoming DORA and NIS2 requirements on third-party cyber risk, have you assessed your cryptographic readiness? Orange Cyberdefense can help you build a migration roadmap to quantum-safe solutions.'
Strategist / Innovator	Commission a deep-dive brief on post-quantum cryptography readiness for EU financial institutions, mapping DORA and NIS2 requirements to Orange's quantum-safe offers and managed services, and identifying gaps in our current capability pool.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-B5F1E5C72CAB	2026-07-30	Springer Science and Busi...	1	Post-Quantum Cryptography Migration Readiness in Global Capability Centres: A Governance Framew...
SIG-57F4F0DAE044	2026-06-18	Cryptography	1	Mythos-Class Frontier Models and the Compression of Post-Quantum Cryptography Migration Timelin...
SIG-C822609B6A07	2026-05-31	openalex.org	1	Large Scale Actualisation of Post-Quantum Cryptography Migration
SIG-13AAE2CD702E	2026-01-19	MDPI AG	1	Cryptographic Asset Discovery and Inventory for Embedded Systems: A Framework for Post-Quantum ...
SIG-353070727CBB	2026-01-01	openalex.org	1	Institutional Approaches to Post-Quantum Cryptography: A Comparative Analysis of Migration Fram...
SIG-5C7568EBD975	2026-01-01	Elsevier BV	1	The Role of the Second Line of Defense in Post-Quantum Cryptography Migration and Cryptographic...
SIG-D45704585F54	2026-01-01	Elsevier BV	1	Towards a Quantum Risk Index: a risk-based framework for prioritizing Post-Quantum Cryptography...
SIG-8BD0B85D3181	2025-12-24	openalex.org	1	Enterprise Migration to Post-Quantum Cryptography: Timeline Analysis and Strategic Frameworks
SIG-ED58CE7F1766	2025-11-24	openalex.org	1	Estimating Migration Timelines for Enterprises Transitioning to Post-Quantum Cryptography: A Re...
SIG-9C40ACE674BC	2026-08-10	Wolters Kluwer	2	What DORA and NIS2 mean for third-party cyber risk management - Wolters Kluwer
SIG-521EDD056481	2026-06-25	R Street Institute	2	Post-Quantum Cryptography Migration in the United States: Managing Risk and Advancing Cyber Rea...
SIG-DD1DB909E321	2026-08-03	arxiv.org	3	Measuring Post-Quantum TLS Deployment Across UK Internet Sectors
SIG-47DFEDD3D7A2	2026-07-15	arxiv.org	3	Multivariate Cryptography-Based Anonymous Certificate Scheme

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (3 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:27+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.